

MULTI-AGENT AI FOR GAME DEVELOPMENT

ASSIGNMENT #8

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NARRATIVE ENGINE PROTOTYPE

(OPTIONAL)

DUE DATE: 25 August 2026, 11:59 PM ET

TIME ESTIMATE: 4–6 hours

PROJECT TYPE: Standalone (does not need to connect to your capstone game, though you may adapt it later if you wish)

OVERVIEW

Objective: Build a virtual Dungeon Master (DM) agent that can roleplay a game world.

Your story engine must dynamically track what a player does—not just what they say—and use that tracked state to generate a reactive, consistent narrative over multiple turns.

CORE REQUIREMENTS

- **JSON Facts Ledger:** Maintain a state tracker (in JSON format) that updates automatically based on the player's actions.
- **Reactive Dialogue & Narration:** The agent's responses must change based on the current state of the ledger. (Example: A player who betrayed an ally receives different dialogue than a player who remained loyal).
- **Turn Consistency:** The agent must maintain consistency across at least 5 player turns. It cannot contradict itself or forget tracked facts during the session.

DELIVERABLES

01. **Agent Code:** Written in Python using the Claude API.
02. **Short ReadMe:** Must include:
 - A brief description of the world you built.
 - An explanation of what your ledger tracks.
 - One specific moment where the agent surprised you during testing.

GRADING RUBRIC (10 POINTS TOTAL)

CRITERION	DESCRIPTION	POINTS
State Tracking	The agent maintains a facts ledger that updates based on player actions. The ledger state must be visible in the output or logs.	4.0
Reactive Dialogue	The agent's responses change based on ledger state, not just the most recent input.	3.0
Consistency	Over 5+ turns, the agent does not contradict itself or forget tracked facts.	2.0
ReadMe	Describes the world, what is tracked, and includes one interesting moment from testing.	1.0

TOTAL POINTS: 10.0